

High-Selective Upgrading of Ethanol to C_{4–10} Alcohols over Hydroxyapatite Catalyst with Superior Basicity

Jia Wang, Wen-Cui Li, Dan-Hui Sun, Lei He, Bai-Chuan Zhou, and An-Hui Lu*

Cite This: *ACS Sustainable Chem. Eng.* 2025, 13, 36–45

Read Online

ACCESS |



Metrics & More



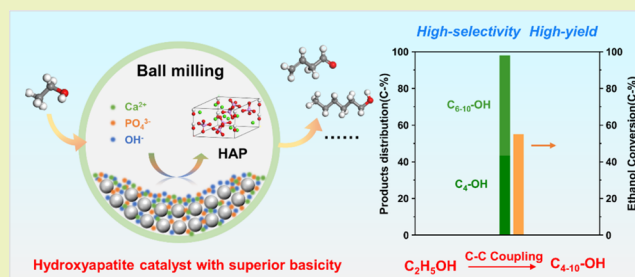
Article Recommendations



Supporting Information

ABSTRACT: The catalytic upgrading of renewable ethanol to C_{4–10} alcohols via C–C coupling offers a green and negative-carbon-emission pathway toward value-added compounds. The manipulation of catalysts' surface basic and acidic properties is the key to achieve high-selectivity C_{4–10} alcohols. In this study, we present a solvent-free mechanochemical approach for the synthesis of hydroxyapatite (HAP) catalysts with enhanced basicity. The selectivity for a total C_{4–10} alcohols reaches 97.8% with a yield of 53.9% at 325 °C and 0.1 MPa, surpassing previously reported catalysts in the literature. The mechanochemically synthesized HAP catalysts extend along the *c*-axis and expose the (002) crystal plane with enriched strong basic [Ca–O–P] sites. CO₂-TPD and XPS analyses demonstrated that the hydrogen bonds between the oxygen atoms of adjoining phosphate groups enhance the basic property of the catalyst surfaces. The kinetic measurements have demonstrated that the abundance of strong basic sites facilitates the adsorption of ethanol molecules and accelerates the rate of C–C coupling reactions, which is responsible for a high yield of C_{4–10} alcohols. This work offers a sustainable approach for synthesizing such alcohols and stimulates the advancement of environmentally friendly catalysts.

KEYWORDS: ethanol, C–C coupling, mechanochemistry, hydroxyapatite, C_{4–10} alcohols



INTRODUCTION

Ethanol produced by biomass fermentation and coal-based energy is a carbon-containing renewable resource and is an essential platform molecule to generate desirable value-added products.^{1,2} C_{4–10} alcohols are significant raw materials for fine chemicals such as surfactants, plasticizers, etc. Moreover, as alcohol-based fuels, high-carbon alcohols exhibit a fuel value comparable to gasoline while possessing a higher air–fuel ratio and greater water compatibility than ethanol. These attributes render them promising biofuels and fuel additives for meeting future demands.³ In comparison to conventional petroleum pathways, the one-pot sustainable process of catalytically converting bioethanol into more profitable C_{4–10} alcohols offers an additional route for reducing carbon emissions, which is consistent with green and ecologically sustainable practices.^{4,5}

Ethanol is upgraded into C_{4–10} alcohols via C–C coupling over acid–base bifunctional catalysts. Several solid catalysts with Lewis acid–base sites, such as hydroxyapatite (HAP, Ca₁₀(PO₄)₆(OH)₂),^{6–8} Mg–Al mixed oxides,^{9–12} and loaded metal oxides,^{13–15} have frequently been used as ethanol catalytic coupling reactions. Typically, the catalytic condensation of ethanol proceeds via the Guerbet mechanism,^{16,17} principally involving dehydrogenation, aldol condensation, dehydration, and hydrogenation to form *n*-butanol.

Among the above-mentioned catalysts, HAP is an attractive and amphoteric material with acid–base pair feature, and their properties are deeply dependent on the preparation method, mainly including liquid-phase^{18,19} and solid-phase methods.^{20,21} The available studies have demonstrated that the catalytic performance is influenced by both the quantity and nature of acid–base active sites present on the surface of hydroxyapatite.²² Currently, the liquid-phase method dominates the modulation of hydroxyapatite surface properties by employing soluble sources of calcium and phosphorus to regulate surface acid–base properties through the addition of modifiers, surfactants, or pH-adjusting agents.^{23,24} However, the wet methods are cumbersome and require the need for bulk solvent for achieving the specific hexagonal phase of HAP. In addition, difficulties in controlling the phase purity and crystallinity of the particles, as well as the complexity and precise controllability of the synthesis process, make some wet methods unsuitable for the large-scale production of HAP.²⁰

Received: May 21, 2024

Revised: December 11, 2024

Accepted: December 12, 2024

Published: December 20, 2024



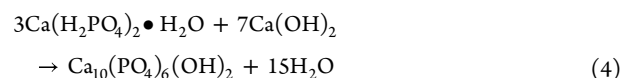
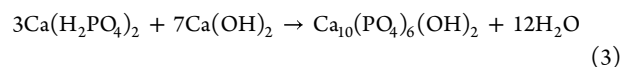
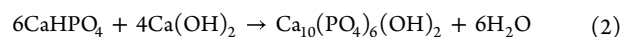
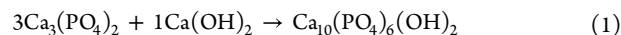
Solvent-free reactions by mechanical action such as ball milling or grinding have been regarded as a promising synthesis method for green chemistry.²⁵ The mechanochemistry approach not only offers the possibility of eliminating the demand for extensive solvent use and reducing waste generation compared to wet methods but also enables to utilize the frictional heat and mechanical momentum generated by the ball milling process to boost intermolecular interactions.^{26,27} Mechanical processing of solids leads to the accumulation of surplus energy, hence emerging surfaces, crystal defects, amorphization, and morphological changes that facilitate the formation of catalytically active sites, which can enhance the reactivity of solid powder.²⁸

In this study, we demonstrated that it is achievable to mechanochemically prepare a variety of hydroxyapatites with different surface acid–base properties and crystallinity by tuning the types and molar ratios of the precursors. The catalyst effectively catalyzed the coupling of ethanol to prepare C_{4–10} alcohols. The geometric and electronic structures and surface properties of HAP catalysts were identified using a variety of characterization tools. The ethanol catalytic conversion pathway was verified via kinetic measurements and in situ spectroscopy experiments. The reaction stability and regeneration performance of ethanol to C_{4–10} alcohols over hydroxyapatite catalysts are discussed.

EXPERIMENTAL SECTION

Catalyst Preparation. The mechanochemical synthesis was carried out in a Fritsch Pulverisette 7 planetary ball mill with 80 mL tanks and a rotation speed of 800 rpm under an ambient atmosphere. In all experiments, the weight ratio of the reactant blends to milling beads was 1:20. The milling beads were zirconia balls with a 5 mm diameter and each mass of about 0.4 g. The duration of the ball milling was 30 min. HAP can be obtained by mechanochemical synthesis containing various mixtures including diverse calcium-containing compounds, phosphates, etc., if the components of the powder mixture are carefully selected based on the chemometric ratio of Ca:P:O:H of the HAP (Scheme 1). The powder mixture was formulated in a suitable molar ratio according to the following chemical reactions 1–4, which were subsequently used to prepare HAP, maintaining a theoretical Ca/P ratio of 1.67. The ball-milled powder samples were calcined at 400 °C with a temperature ramp of 5 °C/min in air for 2 h. These samples are labeled HAP-1, HAP-2,

HAP-3, and HAP-4, respectively. The generation of 1 mol of HAP is accompanied by the formation of different molar amounts of H₂O in each synthesis reaction. Commercial Ca₃(PO₄)₂ calcined at 400 °C were used as representative catalysts for comparison.



Two powders, Ca(H₂PO₄)₂ and Ca(OH)₂ were chosen for the synthesis of HAP as a reaction mixture according to chemical reaction 3, tuning the acid–base properties of the HAP surface by changing the molar ratio of the raw materials. Similarly, the ball-milled powder samples were calcined at 400 °C in air for 2 h. These samples were noted as HAP-3-*x*, where *x* represents the molar ratio of Ca(H₂PO₄)₂ and Ca(OH)₂.

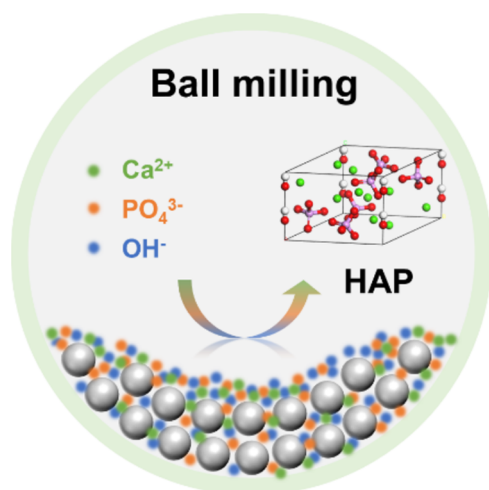
Characterization. Powder X-ray diffraction (XRD) patterns were performed using a Panalytical X'Pert Pro Super X-ray diffractometer with Cu K α radiation (40 kV, 40 mA, λ = 0.15418 nm). N₂ adsorption–desorption isotherms were measured at –196 °C by using a TriStar 3000 physical adsorption analyzer (Micromeritics). The type and number of acid/basic sites of the catalysts were analyzed by NH₃/CO₂-temperature-programmed desorption-mass spectrometry (NH₃/CO₂-TPD-MS) on a Micromeritics AutoChem II 2920 apparatus. X-ray photoelectron spectra (XPS) were recorded with a Thermo Scientific K- α^+ Spectrometer System using a monochromatic Al K α (1486.6 eV) X-ray source. Transmission electron microscopy (TEM) images were obtained with a Tecnai G2 F30 S-Twin instrument operated at 300 kV. Fourier transform infrared spectroscopy (FT-IR) spectra were obtained on a Nicolet 6700 spectrometer. *In situ* ethanol-diffuse reflectance infrared Fourier transform (DRIFT) spectra were captured using a Bruker 70 V spectrometer with a reaction cell (HARRICK) and MCT detector. The time-dependent adsorption profiles of ethanol were recorded on an IGA-100 Intelligent Gravimetric Analyzer (Hidden analytical).

Catalytic Activity Measurements. The ethanol conversion experiments were carried out at atmospheric pressure in a fixed bed reactor with quartz tubes of 8 mm inner diameter. Ethanol (5.7 kPa) diluted with N₂ was flowed through the catalyst bed maintaining a total flow rate of 30 mL·min^{–1}. Before testing, 100 mg of catalyst was pretreated at 400 °C for 2 h in an N₂ atmosphere to remove any H₂O or CO₂ that might have been adsorbed on the catalyst surface. The products were quantitated through online gas chromatography with a flame ionization detector (GC-FID). The connection line between the reaction gas outlet and the chromatography inlet is kept at 240 °C to avoid condensation of the high-carbon compounds generated. The six-way valve outlet of the chromatography is connected to a low-temperature cold trap (0 °C) for collecting the liquid products of the reaction. The identities of products were further confirmed through GC-MS analysis (Agilent 7890A GC interfaced with 5975C MS). The carbon balance of each sample was better than 95%. There are no mass transfer limitations under the test conditions (Figure S1). Details of calculations for ethanol reactivity tests, including ethanol conversion and selectivity, are given in the Supporting Information.

RESULTS AND DISCUSSION

The ethanol coupling activity of HAP catalysts obtained from various feedstocks under identical reaction conditions is shown in Figure 1a and Table S1. The results showed that the HAP-1 catalyst exhibits a low ethanol conversion rate of only 0.2%, with acetaldehyde being the predominant product. The other three catalysts have comparable product distributions, dominated by C_{4–10} alcohols (Table S2). The synthesis of

Scheme 1. Schematic Diagram of the Mechanochemical Synthesis of Hydroxyapatite



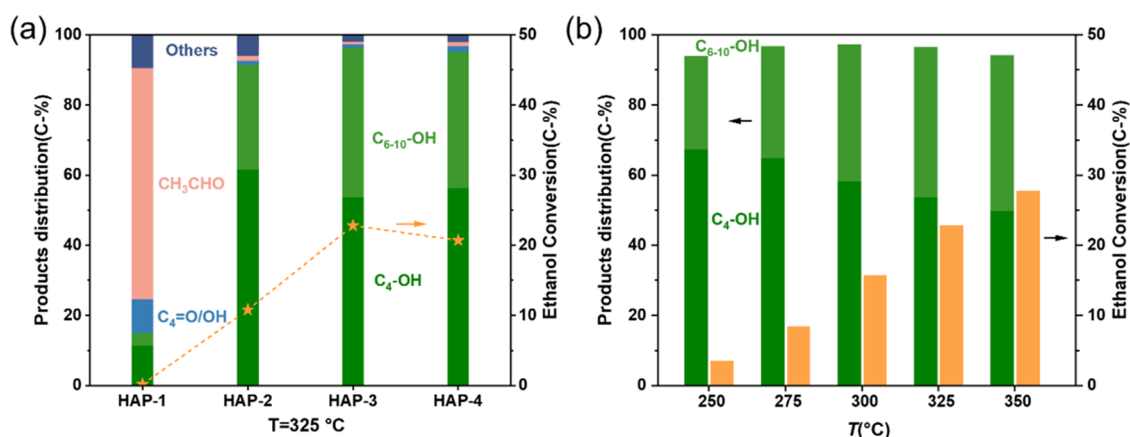


Figure 1. Catalytic performance: (a) Ethanol conversion over HAP catalysts obtained from various feedstock. Reaction activity was measured at 325 °C with WHSV = 1.96 g_{C₂H₅OH} g_{cat}⁻¹·h⁻¹. (b) Ethanol conversion performance over the HAP-3 catalyst at various temperatures. Reaction activity was measured at 250–350 °C with a WHSV of 1.96 g_{C₂H₅OH} g_{cat}⁻¹·h⁻¹.

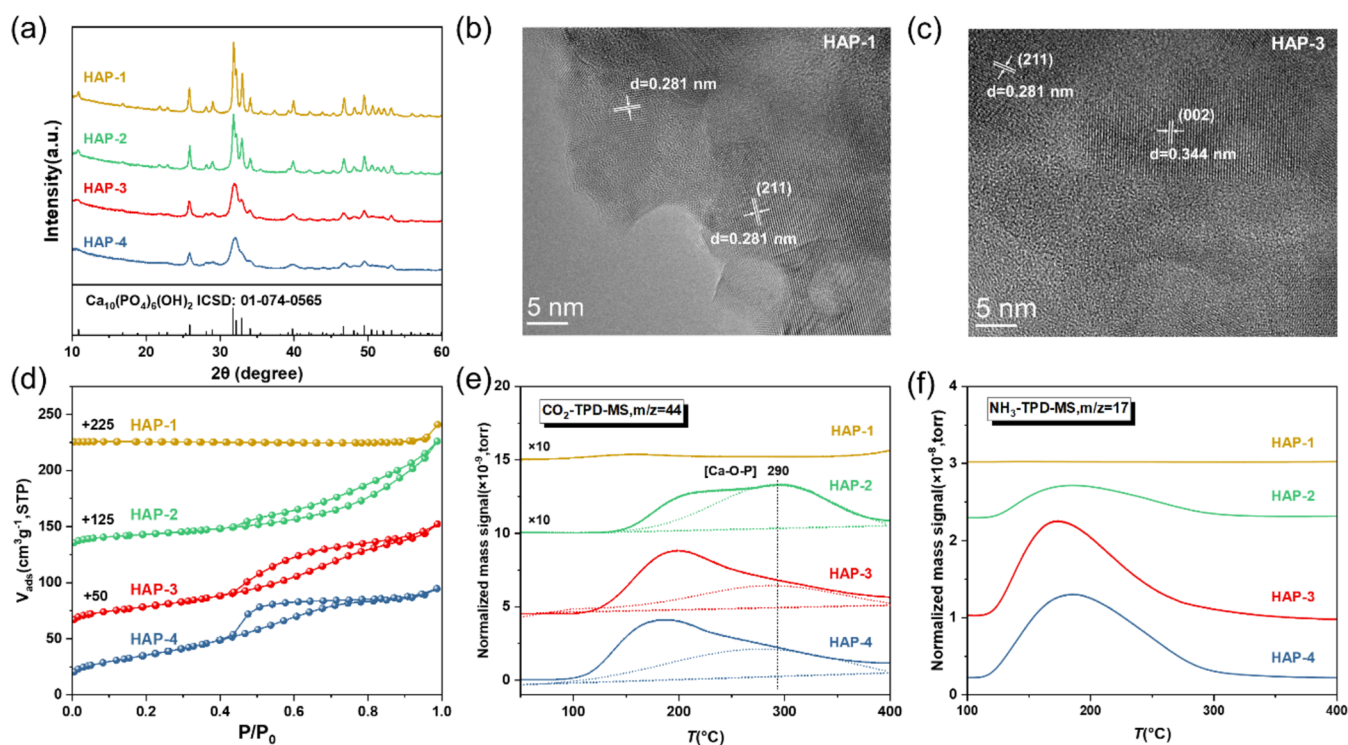


Figure 2. (a) XRD profiles, (b, c) HRTEM images of HAP-1 and HAP-3 catalysts. (d) N₂ adsorption–desorption isotherms, (e) CO₂-TPD-MS signals, and (f) NH₃-TPD-MS signals of HAP catalysts obtained from various feedstock.

HAP is accompanied by the formation of bound water, which affects the surface properties of HAP despite its removal during heat treatment (Table S3). The ethanol conversion on HAP displayed a trend of initial increase, followed by a decrease as the amount of bound water increased. Contrastingly, the HAP-3 catalyst achieved ethanol conversion of 22.0% with 96.1% total C_{4–10} alcohols selectivity at 325 °C.

The ethanol coupling upgrade is a temperature-sensitive reaction that has been investigated for the effect of the reaction temperature on ethanol conversion and product distribution. As shown in Figures 1b and S2, the ethanol conversion increases significantly with raising the reaction temperature. However, more dehydration byproducts such as ethylene, ether, and butadiene are produced when the temperature is increased to 350 °C, causing a decrease in the total C_{4–10}

alcohols selectivity. Impressively, the total selectivity of C_{4–10} alcohols over HAP-3 catalyst was maintained at approximately 95%, especially for the selectivity of C_{6–10} alcohols that increased progressively with increasing temperature.

The different ethanol catalytic activities motivated us to further investigate the structural and surface properties of the catalysts. The physicochemical properties of the HAP catalysts are shown in Figures 2 and S3. Figure 2a provides the crystalline structure of HAP obtained after calcination at 400 °C. The XRD patterns displayed that all materials have major diffraction peaks at 2θ = 25.88, 31.77, and 32.90°, corresponding to the (002), (211), and (300) crystallographic planes of HAP, respectively. The HAP catalysts synthesized mechanochemically exhibit a typical hexagonal phase apatite crystal structure. As shown in Figure S4, the crystalline

Table 1. Physical and Chemical Properties of the Mechanochemically Synthesized HAP Catalysts

catalysts	chemical reaction ^a	molar ratio of $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$	S_{BET} ($\text{m}^2\cdot\text{g}^{-1}$)	Ca/P ratio		acid sites ($\mu\text{mol}\cdot\text{g}^{-1}$) ^c	basic sites ($\mu\text{mol}\cdot\text{g}^{-1}$) ^d		
				theoretical	surface ^b		weak	strong	All
HAP-1	(1)		1.1	1.67		12.6	4.8	0	4.8
HAP-2	(2)		63.4	1.67		73.3	3.0	8.7	10.7
HAP-3	(3)	0.43:1	102.6	1.67	1.48	103.7	19.3	21.0	40.3
HAP-4	(4)		129.3	1.67		106.9	9.3	16.4	25.7
HAP-3-0.40	(3)	0.40:1	117.6	1.75	1.54	63.9	8.1	4.4	12.5
HAP-3-0.42	(3)	0.42:1	119.6	1.69	1.50	81.4	11.2	14.2	25.4
HAP-3-0.44	(3)	0.44:1	115.0	1.64	1.45	107.8	16.9	39.2	56.1
HAP-3-0.45	(3)	0.45:1	113.8	1.61	1.43		8.1	21.7	29.8
HAP-3-0.50	(3)	0.50:1	98.9	1.50			2.94	0	2.94

^aCatalyst synthesis corresponding to chemical reactions 1–4 as described in the Experimental Section. ^bMeasured by XPS. ^cMeasured by NH_3 -TPD. ^dMeasured by CO_2 -TPD.

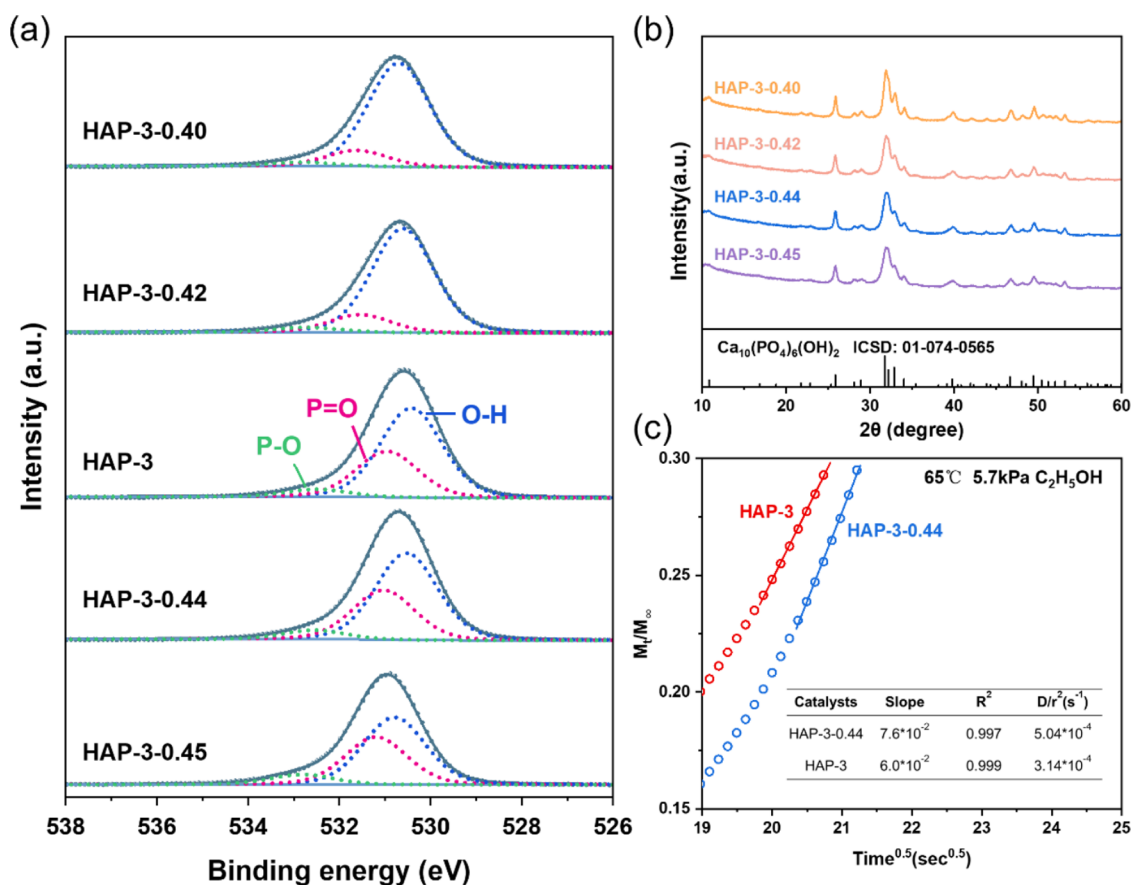


Figure 3. (a) O 1s XPS profile and (b) XRD patterns of catalysts obtained from different molar ratios of $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$. (c) Adsorption kinetic profiles for $\text{C}_2\text{H}_5\text{OH}$ at 65°C over the HAP-3 and HAP-3-0.44 catalysts.

structures of the uncalcined samples and those calcined at 400°C were almost identical, indicating that the bound water accompanying the synthesis of HAP had no effect on the crystalline structures of the samples. Nonetheless, there is a gradual decline in the crystallinity of HAP with increasing amounts of its combined water.²⁹ According to the Scherrer equation, the crystallite size decreased from 43 nm for HAP-1 to 11 nm for HAP-4.

HAP-3 catalyst has a well-defined lattice stripe with a spacing of 0.344 nm (Figures 2c and S5), corresponding to the (002) crystal plane, concluding that the particles extend along the [001] direction of the hexagonal structure, i.e., the *c*-axis.³⁰ The catalyst exposes {010} crystalline facets laterally,³¹ which

are enriched [Ca–O–P] sites owing to the exposure of Ca^{2+} and PO_4^{3-} , whereas the HAP-1 catalyst mainly exposes the (211) crystal plane (Figure 2b). The exposure of the (002) crystal plane necessitates fewer Ca–O bonds to be severed compared to the (211) crystal plane (Figure S6).

The N_2 adsorption–desorption isotherms are shown in Figure 2d. HAP-1 exhibits Type II isotherms belonging to nonporous or macroporous materials with relatively weak interactions between solid surfaces and gas molecules. The other three catalysts show a typical Type IV adsorption isotherm with a hysteresis loop for P/P_0 in the range 0.4–0.95, meaning that the catalysts are mesoporous materials. The HAP-4 catalysts producing the largest amount of combined

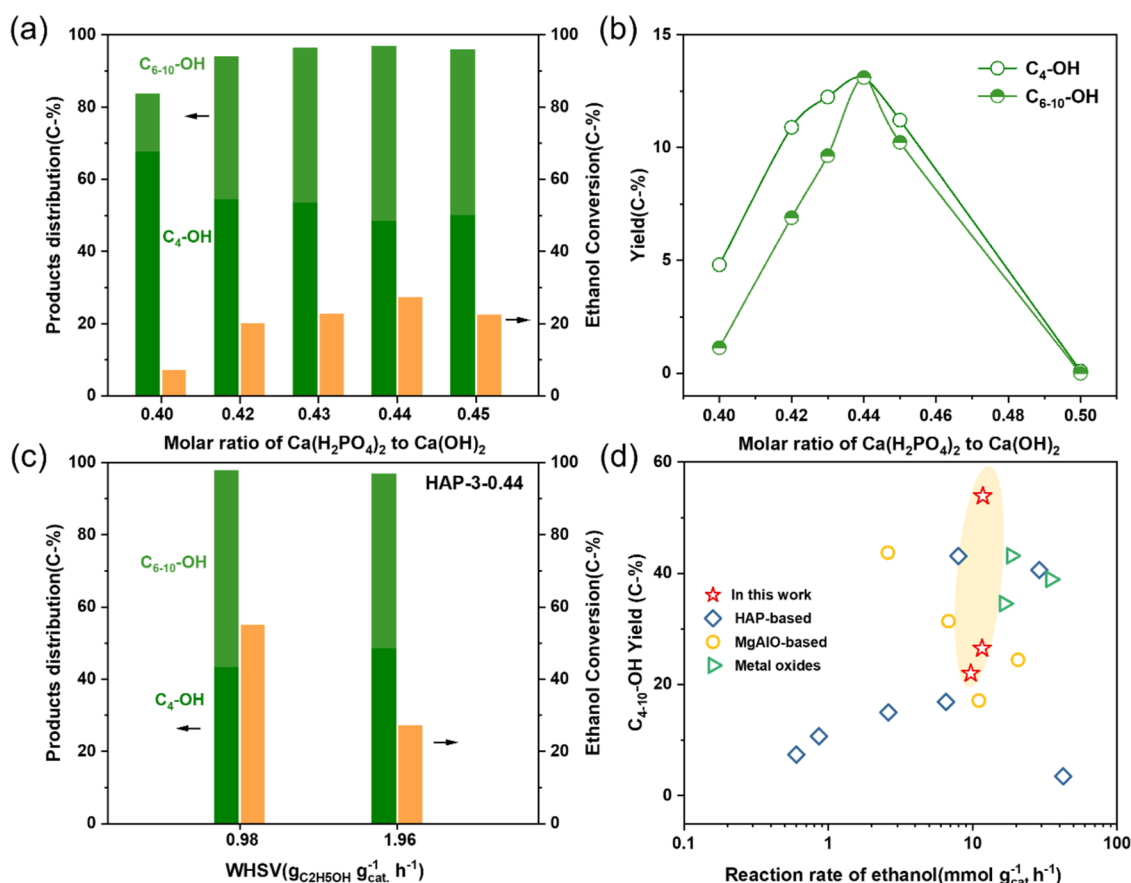


Figure 4. Catalytic performance: (a) Ethanol conversions over HAP catalysts with different Ca(H₂PO₄)₂/Ca(OH)₂ molar ratios. Reaction activity was measured at 325 °C with a WHSV of 1.96 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹. (b) Dependence of high-carbon alcohol yield on the Ca(H₂PO₄)₂/Ca(OH)₂ molar ratio. Reaction activity was measured at 325 °C with a WHSV of 1.96 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹. (c) Ethanol conversion performance over the HAP-3-0.44 catalyst at different WHSV. Reaction activity was measured at 325 °C with a WHSV of 0.98–1.96 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹. (d) Yields of C₄–10 alcohols versus ethanol reaction rate for catalysts described in this work and in the literature (Table S8).

water exhibited the highest specific surface area (Table 1, Figure S7), corresponding to the more disordered HAP crystal structure observed.

Catalysts synthesized from different raw materials were further explored for their surface properties to investigate the influence of surface acidity and basicity on ethanol coupling (Figure 2e,f). Bell and co-workers have observed the existence of weak basic sites ([Ca–O–Ca] or [Ca–OH]) and strong basic sites ([Ca–O–P]) by FTIR of adsorbed CO₂.⁸ The two CO₂ desorption peaks are deconvoluted at 190 and 290 °C, representing the weak and strong basic sites, respectively. HAP-3 catalyst has more [Ca–O–P] strong basic sites attributed to the exposure of the {010} crystal plane (Table 1). In addition, the acidity of these samples was evaluated using NH₃-TPD. It is observed that there is a broad desorption peak over a wide temperature range of 100–300 °C, which indicates the presence of weak-medium acidic sites on the catalyst, and pyridine-IR further reveals that only Lewis acid is exposed on the HAP surface.³² It reinforces that HAP is an acid–base bifunctional catalyst. Notably, catalysts with higher acid/base site ratios lead to additional selectivity of the dehydrated byproduct (Figure S8).

Several investigations have also been conducted to determine how to tailor the surface environments of HAP by varying critical synthesis parameters. By altering the molar ratios of Ca(H₂PO₄)₂/Ca(OH)₂, nonstoichiometric HAP can

be prepared with the bulk theoretical Ca/P ratio of 1.50–1.75. It is established that the surface Ca/P ratio and the atomic coordination environment in general also vary depending on the feedstock ratio, which in turn allows tuning the surface acid–base properties and catalytic performance.³³ All samples exhibit a hexagonal apatite crystalline phase with a consistently high specific surface area of approximately 100 m²/g (Figure 3b).

XPS measurements of the catalysts were carried out to gain further insight into the chemical state of the HAP surface. The full-range scan is presented in Figure S9. Simultaneously, the O 1s XPS survey of these catalysts was examined. The O 1s peak spectrum of HAP was deconvoluted into three peaks,³⁴ including –OH (530.50 eV), P=O (531.08 eV), and P–O (532.48 eV), respectively (Figure 3a, Table S4). It was observed that the total amount of P–O and P=O species (specifically, PO₄^{3–} species) on the sample surface increased with an increasing Ca(H₂PO₄)₂/Ca(OH)₂ ratio. PO₄^{3–} species is the active site responsible for acetaldehyde condensation.⁸ Additionally, XPS analysis of the surface composition indicated a decrease in the Ca/P ratio of the catalyst. For Ca-deficient catalysts, the charge imbalance was revised due to the introduction of H⁺ and the fractional replacement of OH[–] by H₂O. Hydrogen bonds were presented between the oxygens of adjoining phosphate groups.^{35,36} The presence of hydrogen bonds encourages basic sites enhancement. However, when the

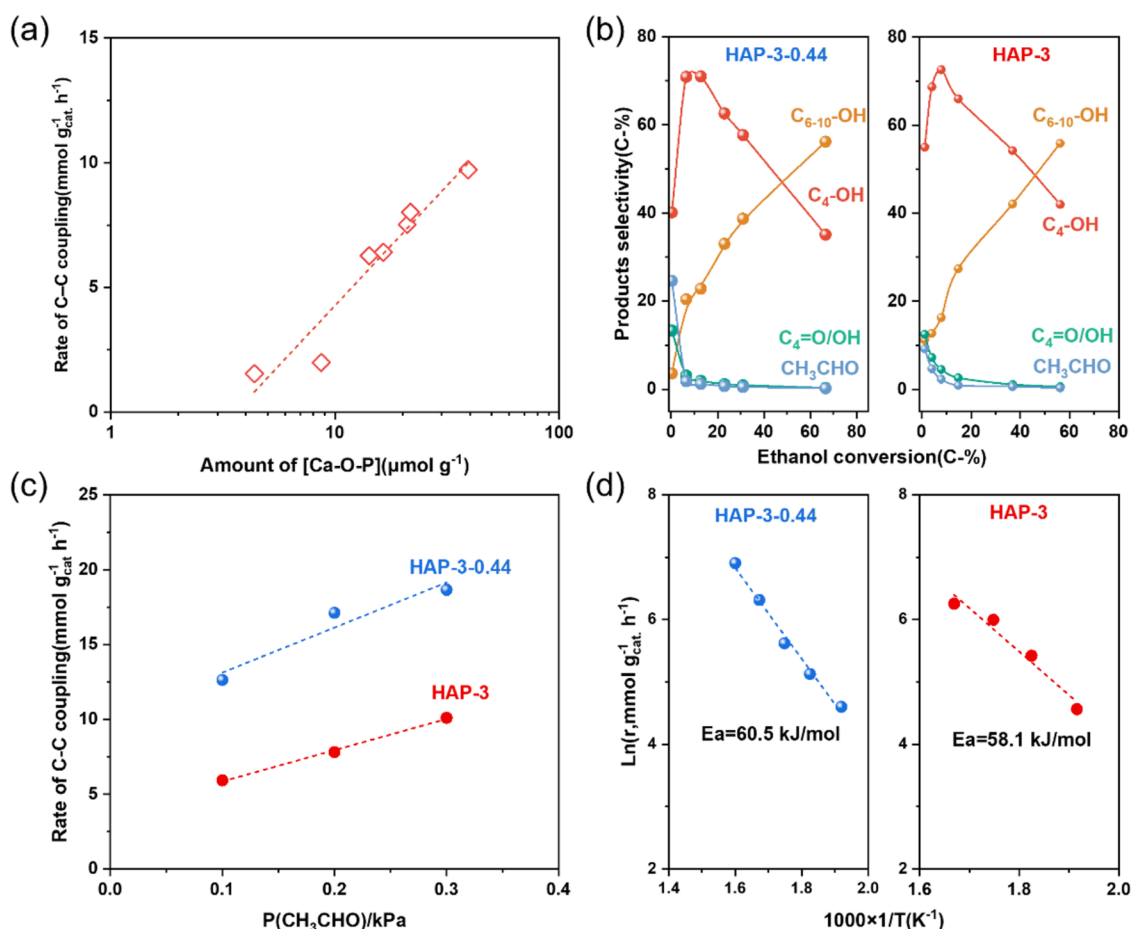


Figure 5. (a) Correlation between the C–C coupling rate and the amount of strong basic $[\text{Ca-O-P}]$ sites. (b) Dependence of the product distributions as a function of ethanol conversion over HAP-3-0.44 and HAP-3 catalysts. Reaction activity was measured at 325 °C with WHSV values of 0.98–40 $\text{g}_{\text{C}_2\text{H}_5\text{OH}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$. (c) Rate of C–C coupling as a function of acetaldehyde partial pressure at constant ethanol pressures. Reaction conditions: $P_{\text{C}_2\text{H}_5\text{OH}} = 5.7 \text{ kPa}$, $T = 325 \text{ }^\circ\text{C}$, HAP mass = 0.05 g, and total flow rate of 30 $\text{mL} \cdot \text{min}^{-1}$. (d) Arrhenius plots of the reaction rate over the HAP catalysts at 250–350 °C.

molar ratio of $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$ exceeds a certain limit, it may have a slight excess of $\text{Ca}(\text{H}_2\text{PO}_4)_2$ attached to the catalyst surface, making the catalyst imperfectly HAP-like in its properties. In contrast, the HAP-3–0.44 catalyst exhibits the highest basic sites among these materials, particularly the amount of strong basic sites of $39.2 \mu\text{mol} \cdot \text{g}^{-1}$ (Table 1). This surpasses the previously reported values of HAP catalysts.

The adsorption of ethanol on the HAP surface constitutes the first step in the coupling reaction. To investigate the diffusion behavior of ethanol in HAP, the time-resolved gas adsorption of ethanol on HAP was measured (Figures 3c and S10, S11). HAP-3-0.44, the more basic sites catalyst, exhibited higher ethanol gas uptake, approximately 1.2 times higher than that of the HAP-3 catalyst. Interestingly, the ethanol diffusional time constant ($D' = D/r^2$, s^{-1}) for HAP-3-0.44 catalyst is 5.04×10^{-4} , which is 1.6 times higher than that of 3.14×10^{-4} for HAP-3 catalyst. As we presumed, the Lewis strong basic sites on the catalyst surface facilitate the adsorption of ethanol molecules for the subsequent C–C coupling reaction.

The HAP surface exposed a suitable surface density of acidic and basic sites which would illustrate the great catalytic activity observed during the reaction (Table S5). As shown in Figure 4a,b, when the $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$ molar ratio was 0.40–0.44, the ethanol conversion increased from 7.1 to 27.3%,

which was improved nearly four times. However, as the $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$ molar ratio increased consistently to 0.45, the ethanol conversion tended to decrease. It obtained the highest C_{4-10} alcohols yield at $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$ molar ratio of 0.44. Meanwhile, the HAP-3-0.44 catalyst achieved a significantly enhanced conversion at a lower WHSV of $0.98 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ while holding a selectivity above 95% for C_{4-10} alcohols (Figures 4c and S12). It obtained a total C_{4-10} alcohols selectivity of 97.8% on the HAP-3–0.44 catalyst with an activity advantage for C_{4-10} alcohols yield (53.9%) compared to other solid acid–base catalysts at an identical reaction rate (ca. $10.0 \text{ mmol} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$) (Figure 4d).

Subsequently, the C–C coupling rate was plotted as a function of the number of $[\text{Ca-O-P}]$ sites. As shown in Figure 5a, there is a linear dependence between the C–C coupling rate and the amount of $[\text{Ca-O-P}]$ sites, indicating that the strong basic sites are responsible for the formation of the C–C bond. This finding confirms that the HAP-3-0.44 catalyst, which contains more strong basic sites, is favorable for ethanol catalytic coupling.

To gain insight into the reaction mechanism and evaluate the effect of different acid/base properties catalysts, kinetic measurements were investigated for the catalytic reactions. The product distribution dependent on ethanol conversion was

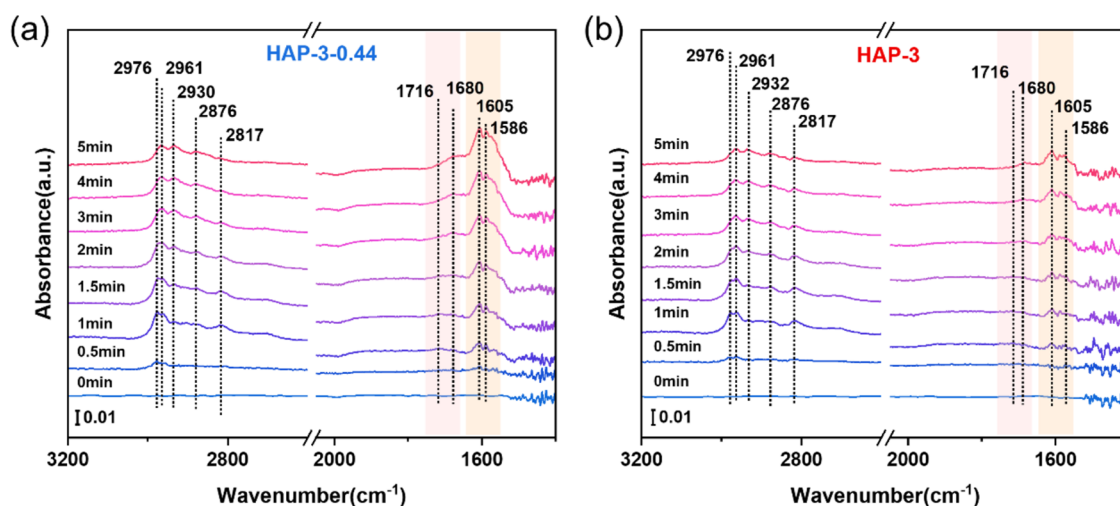


Figure 6. In situ DRIFTS recorded over (a) HAP-3-0.44 and (b) HAP-3 catalyst, both at 325 °C after one dose of ethanol (5×10^{-8} mol) using ~ 40 mg of catalyst.

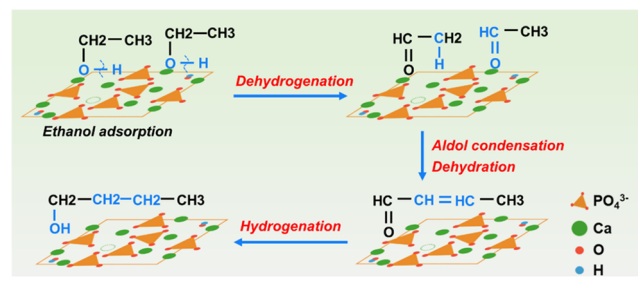
obtained by varying the WHSV (Figure 5b). Acetaldehyde as a primary product gradually decreases with increasing ethanol conversion. The butanol selectivity increases to the highest value of 70% at 10% ethanol conversion. At higher ethanol conversions, the products tend to be C_{6-10} alcohols, indicating that butanol or butenal (C_4 aldehyde intermediate), a butanol-forming precursor, is involved as a secondary product in the formation of higher carbon alcohols. The C_4 aldehyde intermediate can be either hydrogenated to produce butanol or further C–C coupled with aldehyde intermediates to produce heavier alcohols ($C \geq 6$). Aldehyde intermediates generate high-carbon products by stepwise self-coupling or cross-coupling. In addition, the formation rate of butanol and C_{6-10} alcohols reduced in steps (Table S5), illustrating that the mechanism for C–C coupling of ethanol on HAP catalysts adapts to the step-growth model (Figures S13 and S14, Table S6).³⁷ Furthermore, the apparent activation energies on HAP-3-0.44 and HAP-3 catalysts were quite similar, as 60.5 and 58.1 $\text{kJ}\cdot\text{mol}^{-1}$, respectively, meaning that the reaction paths of ethanol on both catalysts were similar (Figure 5d, Table S7).

To accurately assess the role of acetaldehyde in the reaction, ethanol–acetaldehyde reactants were co-fed into the reactor. Figure 5c shows that acetaldehyde exhibits a positive order dependence ($r_{C-C} \sim [\text{CH}_3\text{CHO}]^{0.4 \pm 0.1}$, 0.1–0.3 kPa CH_3CHO , Figure S15). The C–C coupling rate is enhanced by the increase in acetaldehyde partial pressure, suggesting that acetaldehyde is a crucial intermediate in the ethanol coupling reaction. The pathway of ethanol conversion is deduced to follow the “Guerbet” route by the above conclusions.

Furthermore, we investigated possible intermediates for the conversion of ethanol on the catalyst by *in situ* DRIFT experiments.^{38,39} As shown in Figure 6, a slight band (1713 cm^{-1} , corresponding to the $\nu(\text{C}=\text{O})$ mode of acetaldehyde)⁶ was detected within 1 min after a dose of ethanol, which gradually disappeared over time.⁴⁰ There was a gradual increase in the peak associated with butenal at 1680 cm^{-1} . The results implied acetaldehyde and butenal as possible reaction intermediates in the ethanol coupling process.⁴⁰

The kinetic measurements and *in situ* DRIFT experiments demonstrated that acetaldehyde and butenal are crucial intermediates. The ethanol coupling pathway over a strong basic hydroxyapatite catalyst is displayed in Scheme 2, which

Scheme 2. Reaction Mechanism of Ethanol to C_{4-10} Alcohols over the Ball-Milled HAP Catalyst



mainly followed the Guerbet mechanism, this involves the combination of four processes: dehydrogenation of ethanol to acetaldehyde, acetaldehyde to 3-hydroxybutanal via aldol condensation, and further dehydration of 3-hydroxybutanal to butenal (C_4 aldehyde intermediate), followed by hydrogenation of butenal to *n*-butanol. Moreover, the C_4 aldehyde intermediate further self-couples or cross-couples to enable carbon chain growth.

As shown in Figure 7, the ethanol conversion and product distribution dependent on the reaction time of the HAP-3-0.44

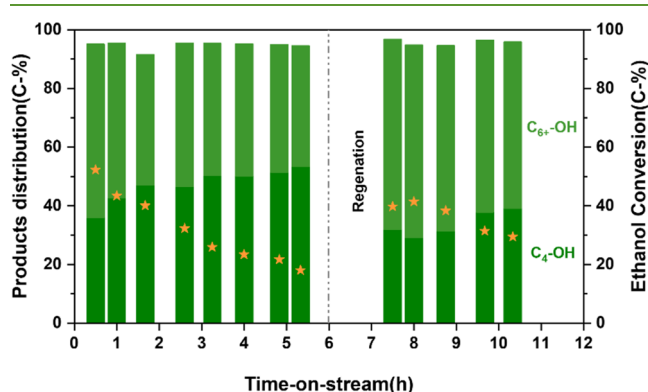


Figure 7. Ethanol conversion and product selectivity as a function of time-on-stream over the HAP-3-0.44 catalyst. Reaction activity was measured at 325 °C in a feed gas of 5.6 vol % $\text{C}_2\text{H}_5\text{OH}/\text{N}_2$ with a WHSV of 0.98 $\text{g}_{\text{C}_2\text{H}_5\text{OH}}/\text{g}_{\text{cat}}\cdot\text{h}^{-1}$.

catalyst were further evaluated. A decrease in the conversion of ethanol from 55 to 43% can be observed during the initial 2 h, but the decline in the conversion slows down after time-on-stream >3 h. Surprisingly, the total alcohol selectivity remains above 95%.

Based on the thermogravimetric (TG) results, the catalyst weight loss begins at 200 °C with a weight loss of about 1.1 wt % (Figure S16). Comparing the Infrared spectra (IR) of fresh and used catalysts, it can be observed that the peaks of $\nu(\text{C}-\text{H})$ for alkanes and $\nu(\text{C}=\text{C})$ for aromatics on the used catalyst. The basicity of the catalyst is key for the C–C coupling reaction. We analyzed the basic sites of fresh, used, and regenerated catalysts using CO_2 -TPD-MS. As shown in Figure S17, the intensity of the CO_2 desorption peak on the used catalyst is weakened, particularly with a significant decrease in the desorption peak corresponding to the Ca–O–P site of the strong basic site (270 °C), indicating a decrease of basic content in the used catalyst. The decrease in the basic site is probably caused by the carbon deposits on the catalyst. After calcination at 450 °C, the catalyst basic sites were re-exposed. The ethanol conversion activity was recovered to be equivalent to the initial 1 h of activity. The recovery of the catalytic activity also demonstrated that the deactivation was caused by carbon deposition covering the active centers. However, while the high-carbon alcohol selectivity can be consistently held above 90%, maintaining high ethanol conversion is still a challenge that needs to be addressed in our future work. The rapid advancement of bioethanol technology suggests that this more efficient and environmentally sustainable process for upgrading bioethanol to higher alcohols holds promising prospects for wider applications.

We conducted a sustainability assessment of our work. Detailed computational analysis is given in the Supporting Information. According to the reaction equations for the ethanol coupling to butanol, hexanol, and octanol, we have calculated the atom economy to be 80.4, 74.0, and 70.7%, respectively. Of these, water is the only byproduct. The conversion of ethanol into higher alcohols exhibits a high atom economy. The HAP-3-0.44 catalyst under reaction conditions of 325 °C and $\text{WHSV} = 1.96 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ was taken as an illustrative example for calculation of the E-factor, PMI, and carbon utilization efficiency. The E-factor of the process is only 0.02. The E-factor gets closer to 0, the more it meets the goal of zero-waste manufacturing. It is worth noting that water does not count as waste in Sheldon's definition of E-factor. The PMI of the process under these conditions is 1.81. We will also prioritize enhancing the ethanol conversion rate to further minimize the RME value. The analysis above implies that the process is compatible with the first principle of Green Chemistry with "Waste prevention not remediation". The total carbon selectivity in byproducts such as ethylene and ether under the reaction conditions is only 3.0%, resulting in an impressive carbon utilization efficiency of up to 97%. Noteworthy, the catalyst preparation and the multiphase reaction do not involve the use of solvents or auxiliaries. Therefore, the conversion of renewable bioethanol to high-carbon alcohols through this route aligns perfectly with the objective of sustainable development.

CONCLUSIONS

In this study, we have prepared a series of catalysts through mechanochemical methods by selecting various powder mixtures based on the chemometric ratio of Ca:P:O:H of the HAP. HAP-3 catalysts obtained from $\text{Ca}(\text{H}_2\text{PO}_4)_2$ and $\text{Ca}(\text{OH})_2$ exhibited great ethanol coupling performance due to the exposure of the (002) crystal plane, which is enriched with [Ca–O–P] sites. It has enabled the controlled preparation of HAP having different surface acid–base properties by tuning the molar ratio of $\text{Ca}(\text{H}_2\text{PO}_4)_2/\text{Ca}(\text{OH})_2$. With an increase in the different ratios, more PO_4^{3-} active sites are available for acetaldehyde condensation. The catalyst with strong basic sites facilitates the adsorption of ethanol molecules so as to accelerate the rate of C–C coupling, following the "Guerbet" pathway thus increasing the yield of C_{4-10} alcohols. The HAP-3-0.44 catalyst exhibited 97.8% selectivity to C_{4-10} aliphatic alcohols with ethanol conversion of 55.1% at 325 °C with a WHSV of $0.98 \text{ g}_{\text{C}_2\text{H}_5\text{OH}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$. The yield of C_{4-10} aliphatic alcohols was thereby close to 54%, which is much greater than previously reported catalysts in the literature. Hence, we believe that this work would provide a viable approach for the green and efficient preparation of highly active ethanol coupling catalysts.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.4c04185>.

Catalyst preparation; characterization methods; catalyst evaluation (Figures S1–S17) (Tables S1–S8) (PDF)

AUTHOR INFORMATION

Corresponding Author

An-Hui Lu – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0003-1294-5928; Email: anhuilu@dlut.edu.cn

Authors

Jia Wang – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Wen-Cui Li – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0001-8066-7144

Dan-Hui Sun – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Lei He – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Bai-Chuan Zhou – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, P. R. China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acssuschemeng.4c04185>

Author Contributions

A.-H.L. conceived and designed the project. J.W. conducted the catalyst preparation, catalytic experiments, and characterization. D.-H.S. and B.-C.Z. assisted in performing kinetic measurements. W.-C.L., L.H., and A.-H.L. have discussed the results and revised the manuscript. All authors have reviewed and contributed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported by the National Key Research and Development Program of China (2021YFA1501301), Dalian Innovation Fund (2021JJ12GX021), and the Fundamental Research Funds for the Central Universities, which are gratefully acknowledged. We are grateful for the support of Liaoning Binhai Laboratory (LBLE-2023-04).

REFERENCES

- (1) Aitchison, H.; Wingad, R. L.; Wass, D. F. Homogeneous Ethanol to Butanol Catalysis—Guerbet Renewed. *ACS Catal.* **2016**, *6* (10), 7125–7132.
- (2) Lin, X.; Fei, X.; Chen, D.; Qi, Y.; Xu, Q.; Liu, Y.; Zhang, Q.; Li, S.; Wang, T.; Qin, Y.; Qiu, X. Efficient Catalytic Upgrading of Ethanol to Higher Alcohols via Inhibiting C–C Cleavage and Promoting C–C Coupling over Biomass-Derived NiZn@NC Catalysts. *ACS Catal.* **2022**, *12* (19), 11573–11585.
- (3) Chen, B.; Zheng, X.; Gu, J.; Qiu, S.; Song, J.; Wu, X.; Dong, H.; Zhang, Q.; Wang, T. Engineering Sn Doping Ni/chitosan to Boost Higher Alcohols Synthesis from Direct Coupling of Aqueous Ethanol: Modifying Adsorption of Aldehyde Intermediates for C–C Bond Cleavage Suppressing. *Appl. Catal., B* **2023**, *321*, No. 121048.
- (4) Eagan, N. M.; Kumbhalkar, M. D.; Buchanan, J. S.; Dumesic, J. A.; Huber, G. W. Chemistries and Processes for the Conversion of Ethanol into Middle-Distillate Fuels. *Nat. Rev. Chem.* **2019**, *3* (4), 223–249.
- (5) Yuan, L.; Zhang, M.; Fan, G.; Li, F. Remarkably Boosting Ethanol Upgrading to Higher Alcohols through Cooperative Catalysis of Surface-Interface Multiple-Active Sites on Nitrogen-Doped Carbon Decorated Copper-Based Catalysts. *Appl. Catal., B* **2024**, *343*, No. 123488.
- (6) Wang, Q. N.; Zhou, B. C.; Weng, X. F.; Lv, S. P.; Schuth, F.; Lu, A. H. Hydroxyapatite Nanowires Rich in [Ca–O–P] Sites for Ethanol Direct Coupling Showing High C_(6–12) Alcohol Yield. *Chem. Commun.* **2019**, *55* (70), 10420–10423.
- (7) Zhou, B.-C.; Li, W.-C.; Lv, W.-L.; Xiang, S.-Y.; Gao, X.-Q.; Lu, A.-H. Enhancing Ethanol Coupling to Produce Higher Alcohols by Tuning H₂ Partial Pressure over a Copper-Hydroxyapatite Catalyst. *ACS Catal.* **2022**, *12* (19), 12045–12054.
- (8) Ho, C. R.; Shylesh, S.; Bell, A. T. Mechanism and Kinetics of Ethanol Coupling to Butanol over Hydroxyapatite. *ACS Catal.* **2016**, *6* (2), 939–948.
- (9) Lv, W.-L.; He, L.; Li, W.-C.; Zhou, B.-C.; Lv, S.-P.; Lu, A.-H. Atomically Dispersed Co²⁺ on MgAlOx Boosting C_{4–10} alcohols Selectivity of Ethanol Valorization. *Green Chem.* **2023**, *25*, 2653–2662.
- (10) Yuan, B.; Zhang, J.; An, Z.; Zhu, Y.; Shu, X.; Song, H.; Xiang, X.; Wang, W.; Jing, Y.; Zheng, L.; He, J. Atomic Ru Catalysis for Ethanol Coupling to C₄₊ Alcohols. *Appl. Catal., B* **2022**, *309*, No. 121271.
- (11) Guo, M. F.; Gray, M. J.; Job, H.; Alvarez-Vasco, C.; Subramaniam, S.; Zhang, X.; Kovarik, L.; Murugesan, V.; Phillips, S.; Ramasamy, K. K. Uncovering the Active Sites and Demonstrating Stable Catalyst for the Cost-Effective Conversion of Ethanol to 1-Butanol. *Green Chem.* **2021**, *23* (20), 8030–8039.
- (12) Lu, B.; Ma, S.; Liang, S.; Wang, Z.; Liu, Y.; Mao, S.; Ban, H.; Wang, L.; Wang, Y. Efficient Conversion of Ethanol to 1-Butanol over Adjacent Acid–Base Dual Sites via Enhanced C–H Activation. *ACS Catal.* **2023**, *13*, 4866–4872.
- (13) Jiang, D.; Fang, G.; Tong, Y.; Wu, X.; Wang, Y.; Hong, D.; Leng, W.; Liang, Z.; Tu, P.; Liu, L.; Xu, K.; Ni, J.; Li, X. Multifunctional Pd@UiO-66 Catalysts for Continuous Catalytic Upgrading of Ethanol to n-Butanol. *ACS Catal.* **2018**, *8* (12), 11973–11978.
- (14) Wang, Z.; Pang, J.; Song, L.; Li, X.; Yuan, Q.; Li, X.; Liu, S.; Zheng, M. Conversion of Ethanol to n-Butanol over NiCeO₂ Based Catalysts: Effects of Metal Dispersion and NiCe Interactions. *Ind. Eng. Chem. Res.* **2020**, *59* (51), 22057–22067.
- (15) He, J.; Li, X.; Kou, J.; Tao, T.; Shen, X.; Jiang, D.; Lin, L.; Li, X. Catalytic Upgrading of Ethanol to Higher Alcohols over Nickel-Modified Cu–La₂O₃/Al₂O₃ Catalysts. *Catal. Sci. Technol.* **2023**, *13* (1), 170–177.
- (16) Kozłowski, J. T.; Davis, R. J. Heterogeneous Catalysts for the Guerbet Coupling of Alcohols. *ACS Catal.* **2013**, *3* (7), 1588–1600.
- (17) Young, Z. D.; Davis, R. J. Hydrogen Transfer Reactions Relevant to Guerbet Coupling of Alcohols over Hydroxyapatite and Magnesium Oxide Catalysts. *Catal. Sci. Technol.* **2018**, *8* (6), 1722–1729.
- (18) Szenknect, S.; Mesbah, A.; Descostes, M.; Maihachi-Ahmed, A.; Bonato, L.; Massonnet, M.; Ziouane, Y.; Vors, E.; Vercouter, T.; Clavier, N.; Lautru, J.; Dacheux, N. Uranium Removal from Mining Water using Cu Substituted Hydroxyapatite. *J. Hazard. Mater.* **2020**, *392*, No. 122501.
- (19) Um, S. H.; Chung, Y. W.; Seo, Y.; Seo, H.; Ok, M. R.; Kim, Y. C.; Han, H. S.; Chung, J. J.; Edwards, J. R.; Jeon, H. Robust Hydroxyapatite Coating by Laser-Induced Hydrothermal Synthesis. *Adv. Funct. Mater.* **2020**, *30* (48), No. 2005233, DOI: 10.1002/adfm.202005233.
- (20) Fihri, A.; Len, C.; Varma, R. S.; Solhy, A. Hydroxyapatite: A Review of Syntheses, Structure and Applications in Heterogeneous Catalysis. *Coord. Chem. Rev.* **2017**, *347*, 48–76.
- (21) Yook, H.; Hwang, J.; Yeo, W.; Bang, J.; Kim, J.; Kim, T. Y.; Choi, J. S.; Han, J. W. Design Strategies for Hydroxyapatite-Based Materials to Enhance Their Catalytic Performance and Applicability. *Adv. Mater.* **2023**, *35*, No. 2204938.
- (22) Ogo, S.; Onda, A.; Yanagisawa, K. Selective Synthesis of 1-Butanol from Ethanol over Strontium Phosphate Hydroxyapatite Catalysts. *Appl. Catal., A* **2011**, *402* (1–2), 188–195.
- (23) Tsuchida, T.; Kubo, J.; Yoshioka, T.; Sakuma, S.; Takeguchi, T.; Ueda, W. Reaction of Ethanol over Hydroxyapatite Affected by Ca/P Ratio of Catalyst. *J. Catal.* **2008**, *259* (2), 183–189.
- (24) Ogo, S.; Onda, A.; Iwasa, Y.; Hara, K.; Fukuoka, A.; Yanagisawa, K. 1-Butanol Synthesis from Ethanol over Strontium Phosphate Hydroxyapatite Catalysts with Various Sr/P ratios. *J. Catal.* **2012**, *296*, 24–30.
- (25) Chaikina, M. V.; Bulina, N. V.; Vinokurova, O. B.; Prosanov, I. Y.; Dudina, D. V. Interaction of Calcium Phosphates with Calcium Oxide or Calcium Hydroxide during the “Soft” Mechanochemical Synthesis of Hydroxyapatite. *Ceram. Int.* **2019**, *45* (14), 16927–16933.
- (26) Eckert, R.; Felderhoff, M.; Schuth, F. Preferential Carbon Monoxide Oxidation over Copper-Based Catalysts under In Situ Ball Milling. *Angew. Chem., Int. Ed.* **2017**, *56* (9), 2445–2448.
- (27) Dinda, S.; Bhagavatam, A.; Alrehaili, H.; Dinda, G. P. Mechanochemical Synthesis of Nanocrystalline Hydroxyapatite from Ca(H₂PO₄)₂·H₂O, CaO, Ca(OH)₂, and P₂O₅ Mixtures. *Nanomaterials* **2020**, *10* (11), No. 2232.
- (28) Friščić, T.; Mottillo, C.; Titi, H. M. Mechanochemistry for Synthesis. *Angew. Chem., Int. Ed.* **2020**, *59* (3), 1018–1029.
- (29) Corno, M.; Busco, C.; Bolis, V.; Tosoni, S.; Ugliengo, P. Water Adsorption on the Stoichiometric (001) and (010) Surfaces of

Hydroxyapatite: A Periodic B3LYP Study. *Langmuir* **2009**, *25* (4), 2188–2198.

(30) Oh, S. C.; Xu, J.; Tran, D. T.; Liu, B.; Liu, D. Effects of Controlled Crystalline Surface of Hydroxyapatite on Methane Oxidation Reactions. *ACS Catal.* **2018**, *8* (5), 4493–4507.

(31) Ospina, C. A.; Terra, J.; Ramirez, A. J.; Farina, M.; Ellis, D. E.; Rossi, A. M. Experimental Evidence and Structural Modeling of Nonstoichiometric (010) Surfaces Coexisting in Hydroxyapatite Nano-Crystals. *Colloids Surf., B* **2012**, *89*, 15–22.

(32) Zikrata, O. V.; Larina, O. V.; Valihura, K. V.; Kyriienko, P. I.; Balakin, D. Y.; Khalakhan, I.; Veltruská, K.; Krajnc, A.; Mali, G.; Soloviev, S. O.; Orlyk, S. M. Successive Vapor-Phase Guerbet Condensation of Ethanol and 1-Butanol to 2-Ethyl-1-hexanol over Hydroxyapatite Catalysts in a Flow Reactor. *ACS Sustainable Chem. Eng.* **2021**, *9* (51), 17289–17300.

(33) Kiani, D.; Baltrusaitis, J. Surface Chemistry of Hydroxyapatite for Sustainable n-Butanol Production from Bio-Ethanol. *Chem. Catal.* **2021**, *1* (4), 782–801.

(34) Liu, X.; Yin, H.; Liu, H.; Cai, Y.; Qi, X.; Dang, Z. Multicomponent Adsorption of Heavy Metals onto Biogenic Hydroxyapatite: Surface Functional Groups and Inorganic Mineral Facilitating Stable Adsorption of Pb(II). *J. Hazard. Mater.* **2023**, *443* (Pt A), No. 130167.

(35) Rodrigues, E. G.; Keller, T. C.; Mitchell, S.; Pérez-Ramírez, J. Hydroxyapatite, an Exceptional Catalyst for the Gas-Phase Deoxygenation of Bio-Oil by Aldol Condensation. *Green Chem.* **2014**, *16* (12), 4870–4874.

(36) Posner, A. S.; Stutman, J. M.; Lippincott, E. R. Hydrogen-Bonding in Calcium-Deficient Hydroxyapatites. *Nature* **1960**, *188*, 486–487.

(37) Moteki, T.; Flaherty, D. W. Mechanistic Insight to C–C Bond Formation and Predictive Models for Cascade Reactions among Alcohols on Ca- and Sr-Hydroxyapatites. *ACS Catal.* **2016**, *6* (7), 4170–4183.

(38) Ramesh, K.; Ling, E. G. Y.; Gwie, C. G.; White, T. J.; Borgna, A. Structure and Surface Reactivity of WO_4^{2-} , SO_4^{2-} , PO_4^{3-} Modified Ca-Hydroxyapatite Catalysts and Their Activity in Ethanol Conversion. *J. Phys. Chem. C* **2012**, *116* (35), 18736–18745.

(39) Hanspal, S.; Young, Z. D.; Shou, H.; Davis, R. J. Multiproduct Steady-State Isotopic Transient Kinetic Analysis of the Ethanol Coupling Reaction over Hydroxyapatite and Magnesia. *ACS Catal.* **2015**, *5* (3), 1737–1746.

(40) Wang, Q.-N.; Weng, X.-F.; Zhou, B.-C.; Lv, S.-P.; Miao, S.; Zhang, D.; Han, Y.; Scott, S. L.; Schüth, F.; Lu, A.-H. Direct, Selective Production of Aromatic Alcohols from Ethanol Using a Tailored Bifunctional Cobalt–Hydroxyapatite Catalyst. *ACS Catal.* **2019**, *9* (8), 7204–7216.